

# Lesson 20 ORDINARY DIFFERENTIAL EQUATIONS (ODEs)

## DIFFERENTIAL EQUATION

Defn: A differential equation is an equation that involves independent variable  $x$ , dependent variable  $y$  and one or more derivatives w.r.t  $x$ .

### Example

(i)  $x \frac{dy}{dx} + y = e^x$  [1<sup>st</sup> order D.E]

(ii)  $x^2 \frac{d^2y}{dx^2} - xy \frac{dy}{dx} = 3x^3$  [2<sup>nd</sup> order D.E]

(iii)  $3x \frac{d^3y}{dx^3} + 3y = e^x \frac{dy}{dx}$  [3<sup>rd</sup> order D.E]

### Defn: (Order of D.E)

The order of a differential equation is given by the highest derivative involved in the equation.

## SOLVING FIRST ORDER DIFFERENTIAL EQUATION

### Method 1: DIRECT INTEGRATION

If the equation can be arranged in the form  $\frac{dy}{dx} = f(x)$ , then the equation can be solved by direct or simple integration.

### Example 1

Solve the following differential equation:

(a)  $\frac{dy}{dx} = 3x^2 - 6x + 5$

(b)  $e^x \frac{dy}{dx} = 4$ ,  $y(0) = 3$

(c)  $x \frac{dy}{dx} = 5x^3 + 4$

## Solution

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$$(a) \frac{dy}{dx} = 3x^2 - 6x + 5$$

$$y = \int (3x^2 - 6x + 5) dx$$

$$\underline{\underline{y = x^3 - 3x^2 + 5x + C}}, \text{ where } C - \text{arbitrary Constant.}$$

$$(b) e^x \frac{dy}{dx} = 4, \quad y(0) = 3.$$

$$\frac{dy}{dx} = 4e^{-x}$$

$$y = 4 \int e^{-x} dx = -4e^{-x} + C$$

$$y(0) = 3 \xrightarrow{\text{Means}} \begin{cases} x=0 \\ y=3 \end{cases}$$

$$y = -4e^{-x} + C$$

$y(0) = 3$  means  
put  $x=0, y=3$

$$3 = -4 + C$$

$$\therefore C = 7$$

$$\underline{\underline{y = -4e^{-x} + 7}}$$

$$(c) x \frac{dy}{dx} = 5x^3 + 4$$

$$\frac{dy}{dx} = 5x^2 + \frac{4}{x}$$

$$y = 5 \int x^2 dx + 4 \int \frac{1}{x} dx$$

$$\underline{\underline{y = \frac{5}{3}x^3 + 4 \ln x + C}}$$

## Method 2: SEPARATION OF VARIABLES

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If the differential equation is of the form

$$\frac{dy}{dx} = f(x, y)$$

and  $f(x, y)$  is separable i.e.  $f(x, y) = g(x)h(y)$

then

$$\frac{dy}{dx} = g(x)h(y)$$

$\left( \begin{array}{c} \text{y only} \end{array} \right) \rightarrow \int \frac{1}{h(y)} dy = \int g(x) dx \leftarrow \left( \begin{array}{c} \text{x only} \end{array} \right)$

### Example 1

Solve the following differential equation:

(a)  $\frac{dy}{dx} = \frac{2x}{y+1}$

(b)  $\frac{dy}{dx} = (1+x)(1+y)$

(c)  $\frac{dy}{dx} = \frac{y^2 + xy^2}{x^2y - x^2}$

### Solution

(a)  $\frac{dy}{dx} = \frac{2x}{y+1}$

$$\int (y+1) dy = 2 \int x dx$$

$$\underline{\underline{\frac{1}{2}y^2 + y = x^2 + C}}$$

(b)  $\frac{dy}{dx} = (1+x)(1+y)$

$$\int \frac{1}{1+y} dy = \int (1+x) dx$$

$$\underline{\underline{\ln(1+y) = x + \frac{1}{2}x^2 + C}}$$

(c)  $\frac{dy}{dx} = \frac{y^2 + xy^2}{x^2y - x^2} = \frac{y^2(1+x)}{x^2(y-1)}$

$$\int \frac{y-1}{y^2} dy = \int \frac{1+x}{x^2} dx$$

$$\int \left( \frac{1}{y} - y^{-2} \right) dy = \int \left( x^{-2} + \frac{1}{x} \right) dx$$

$$\ln y + \frac{1}{y} = -\frac{1}{x} + \ln x + C$$



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$$(a) \quad xy \frac{dy}{dx} = \frac{x^2 + 1}{y + 1}$$

$$(b) \quad y \tan x \frac{dy}{dx} = (4 + y^2) \sec^2 x$$

$$(c) \frac{\sin x}{1+y} \frac{dy}{dx} = \cos x$$

$$(a) \quad xy \frac{dy}{dx} = \frac{x^2 + 1}{y + 1}$$

$$\int \frac{y}{y+1} dy = \int \frac{x^2+1}{x} dx$$

$$\int \left(1 - \frac{1}{y+1}\right) dy = \int \left(x + \frac{1}{x}\right) dx$$

$$y - \ln(y+1) = \frac{1}{2}x^2 + \ln x + C$$

$$(c) \frac{\sin x}{1+y} \frac{dy}{dx} = \cos x$$

$$\int \frac{1}{1+y} dy = \int \frac{\cos x}{\sin x} dx$$

$$\ln(1+y) = \ln(\sin x) + \ln A$$
$$= \ln(A \sin x)$$

ie  $1+y = A \sin x$

$$y = A \sin x - 1$$

$$(b) \quad y \tan x \frac{dy}{dx} = (4 + y^2) \sec^2 x$$

$$\frac{1}{2} \int \frac{2y}{4+y^2} dy = \int \frac{\sec^2 x}{\tan x} dx$$

$$\frac{1}{2} \ln(4+y^2) = \ln(\tan x) + \ln A = \ln(A \tan x)$$

$$\ln(\sqrt{4+y^2}) = \ln(A \tan x)$$

$$\sqrt{4+y^2} = A \tan x$$